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INHIBITORY EFFECT OF HERBAL REMEDIES ON 12-O-TETRADECANOYLPHORBOL-13-ACETATE-PROMOTED EPSTEIN–BARR VIRUS EARLY ANTIGEN ACTIVATION†

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For the past several years we have been evaluating natural products as potential cancer chemopreventive agents in a short term *in vitro* assay involving Epstein–Barr virus early antigen (EBV-EA) activation in Raji cells promoted by phorbol ester, 12-O-tetradecanoylphorbol-13-acetate (TPA). Because of the current interest in the use of herbal remedies, we considered examining them for their cancer chemopreventive activities, using their extracts with a view to uncovering such benefits (if any) these remedies might possess. Thirty-six extracts of 32 herbs belonging to 27 families in use as herbal remedies including those of gingko, black cohosh, echinacea, kava-kava, saw palmetto, turmeric, angelica, wild yam, cat's claw, passion flower, muira puama, feverfew, blueberry, chasteberry, licorice, nettle, golden seal, pygeum, ginger, valerian and hops were prepared and evaluated. Turmeric at a concentration of 10 μg ml^{-1} exhibited the most potent anti-EBV-EA activity, which is ten times more than passionflower, that is next in the order of activity. At the concentration level of 100 μg ml^{-1}, several of the herbal remedies tested inhibited the EBV-EA in Raji cells exposed to the tumor promoter TPA (32 pM) by more than 90%. We also report for the first time the activities of 16 new medicinal plants as potential cancer chemopreventive agents. Since inhibitors of EBV-EA promoted by TPA *in vitro* have been shown to be effective anti-tumor promoting agents in laboratory animal models, our results indicate new and potential applications of these herbal remedies as cancer chemopreventive agents since they are already in clinical use in the human population.

KEY WORDS: herbs, medicinal plants, cancer, chemoprevention, Epstein–Barr virus.

INTRODUCTION

The disease-oriented screening of traditional remedies is now recognized as an essential tool for new drug discovery. Medicinal plants and/or herbal remedies derived from these studies are being increasingly used by the general public as 'health food/dietary supplements' on a self-selection basis to either replace or complement conventional medicines. This increased utilization in part may be due to the general disillusionment with conventional medicines coupled with the desire for a 'natural' life style. In addition, a lot of scientific evidence involving both basic scientific research and case-controlled clinical trials have validated the use of these herbal remedies in the prevention and/or treatment of a variety of human ailments. One of the top selling herbs, echinacea, is a widely used phytomedicinal for the treatment of the common cold and upper respiratory tract infections. Twelve clinical studies published from 1961 to 1997 concluded that echinacea was efficacious in treating the common cold [1]. Similar proof of efficacy has been established for turmeric in the treatment of biliary dyskinesia [2], valerian in the treatment of sleep disorders [3] and St. John's wort in the treatment of mild to moderate depression [4]. In patients suffering from peripheral occlusive arterial disease in Fontaine stage IIb, treatment with ginkgo was found to be very safe and therapeutically relevant [5]. Ginkgo was also effective in the treatment of Alzheimer's diseases [6].

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For several years, we have been interested in the study of natural products, particularly their ability to function as cancer chemopreventive agents [7, 8]. We have demonstrated the anti-tumor promoting activity of natural colorants, vegetables, fruits and other plant parts [9–11] which are in use as foods, pharmaceuticals and cosmetic preparations. It is our hypothesis that 'it is easier to achieve acceptance of popular herbal remedies already in use for an additional therapeutic gain than to introduce a new product'. Based on this, we have studied the cancer chemopreventive potentials of 36 extracts of 32 herbs belonging to 27 families in use as herbal remedies in different parts of the world. The selected herbs are in clinical use for various medical ailments (Table I) including malignant conditions [12–17]. Earlier, we have shown a correlation between the *in vitro* anti-tumor promoting effects on Epstein–Barr virus early antigen (EBV-EA) activation induced by the tumor promoter 12-O-tetradecanoylphorbol-13-acetate (TPA) and specific *in vivo* chemopreventive effect in the tumor promotion stage of carcinogenesis in the skin and lung [8–10]. The current paper describes the studies we have undertaken to define additional medical benefits (if any) of these herbs as potential cancer chemopreventive agents.

MATERIALS AND METHODS

Chemicals

All chemicals used were of the highest purity commercially available. TPA was obtained from Wako Pure Chemical Industries, Osaka, Japan. Guggulipid, an extract of oleoresin of *Commiphora mukul*, and boswellin, an extract of oleoresin of *Boswellia serrata*, were procured from Sabinsa Corp., Piscataway, NJ, USA. Commercial extract of *Curcuma longa* was obtained from Kalsec, Kalamazoo, MI, USA. All other herbal products were purchased from Penn Herb Co. Ltd., Philadelphia, PA, USA.

Preparation of the Herbal Extracts

Powdered plant material (5 g) of the herbal remedies was stirred for 2 h at room temperature with 100 ml of 95% ethanol. The mixture was centrifuged for 10 min and the supernatant decanted and filtered with a suction pump. The residue was washed twice with a small volume of 95% ethanol and filtered. The filtered extracts and washings were combined and evaporated to dryness under reduced pressure using a rotary evaporator. The residual extract left was dried under vacuum and weighed. The yields of the ethanol extractions are recorded in Table II. In the case of guggulipid, the commercial product was extracted first with hexane. The hexane-insoluble material was then extracted in sequence with ethyl acetate and methanol. The yields of the three fractions were also recorded in Table II. Boswellin, a commercial extract of *Boswellia serrata* and turmeric, the commercial extract of *Curcuma longa* were tested without further processing. In all the cases, samples of the crude extracts were retained for future reference.

Evaluation of EBV-EA activation

The inhibition of EBV-EA activation was assayed as previously described [18, 19]. In the present study, the Epstein–Barr virus activation was assayed using the EBV genome carrying lymphoblastoid Raji cells, which were cultivated in RPMI 1640 medium. The cells were incubated at 37 °C for 48 h in 1 ml of the medium containing butyric acid (4 mM, co-inducer), TPA (32 pM) and known amount of test compound in dimethylsulfoxide (DMSO). Smears were made from cell suspension. The EBV-EA expressing cells were stained with high titer EBV-EA positive serum from nasopharyngeal carcinoma (NPC) patients and detected by an indirect immunofluorescence technique. In each assay 500 cells were counted and the number of stained cells (positive cells) among them was recorded. The experiments were done in triplicate for each data point and the results are means of at least three independent experiments. All the extracts were tested at three dose levels and the results are presented as mean $\pm$ SEM. The EBV-EA inhibitory activity of the test extract was compared with that of a positive control experiment (100%) with butyric acid (4 mM) and TPA (32 pM). In the experiments, the EBV-EA induction was ordinarily around 30%.

Evaluation of cell viability

The cytotoxicity or cell viability of surviving cells was determined using the trypan-blue staining method. After EBV-EA activation assays, 0.1 ml of treated cells (reacted cells) were suspended in phosphate-buffered saline (PBS) and stained with 0.1 ml of 0.25% trypan-blue solution. Dying cells were dyed blue and the non-blue-dyed cells were counted. It was essential to have cell viability of more than 60% for accessing the anti-tumor promoting activity of the test substance [20].

Data analysis

The statistical analysis of these data was performed using the Student t-test.

RESULTS

Table II shows the yields of the extracts from the herbal remedies. The results of the study showed that the 36 herbal extracts tested exhibited statistically significant inhibitory effect on the experimental EBV-EA activation promoted by TPA as shown in Table II. Co-administration of TPA at 32 pM concentration showed 100% induction of the EBV-EA. However, the addition of the extracts of the herbal recipes exhibited significant inhibition of the EBV-EA to varying degrees. Most of the extracts exhibited inhibitory effect in a dose-dependent manner.

Table I
Main clinical indications of the medicinal plants

No.	*Botanical names (family)*	*Common name*	*Plant part consumed*	*Clinical indications*
1.	*Angelica archangelica* L.* (Apiaceae)	Angelica	Seed, whole herb, root, rhizome	Used as a diuretic, diaphoretic, and for the treatment of loss of appetite, peptic discomforts, feeling of fullness, flatulence and anorexia nervosa
2.	*Asparagus officinalis* L. (Liliaceae)	Asparagus	Whole herb, rhizome, roots	Used for irrigation therapy of the urinary tract and for prevention of kidney stones
3.	*Boswellia serrata* Roxb (Burseraceae)	Boswellin	Bark, resin	Treatment of chronic lung diseases, diarrhea, dysentery, pulmonary disease, menorrhea, dysmenorrhea, gonorrhea, syphilitic affection, piles, liver disorders and ringworm
4.	*Caulophyllum thalictroides* (L.) Michx (Berberidaceae)	Blue cohosh	Rhizome, root	Used as antispasmodic for menstrual ailments, cramps, and gynecological disorders. Worm infestations and dehydration
5.	*Cimicifuga racemosa* (L.) Nutt.* (Ranunculaceae)	Black cohosh	Rhizome, root	Premenstrual discomforts and dysmenorrhea or climacteric (menstrual) neuro-vegetative ailments
6.	*Commiphora mukul* (Hooker, Stedor) Engl. (Burseraceae)	Guggulipid	Gum-resin	Lowering of blood cholesterol
7.	*Curcuma longa* L.* (Zingiberaceae)	Turmeric	Rhizome	Treatment of acid, flatulent,or atonic dysepsia. Also for the treatment of cancer, jaundice, liver, malaria, small pox, chicken pox and as an antiseptic
8.	*Dioscorea villosa* L. (Dioscoreaceae)	Wild yam	Rhizome, root	Natural alternative to estrogen replacement therapy
9.	*Echinacea purpurea* (L.) Moench*$ (Asteraceae)	Echinacea	Rhizome, root	Immunostimulant, broad spectrum antimicrobial, inflammation of the mouth and pharynx, topical anti-infective, arthritis, common cold and flu, cough and bronchitis, fevers and cold, and urinary tract infections
10.	*Ginkgo biloba* L.*$ (Ginkgoaceae)	Gingko	Leaf, seed	Vascular insufficiency, depression, intermittent claudication, Alzheimer's disease, memory enhancement, asthma, vertigo, headache and tinnitus, senility and diabetic retinopathy
11.	*Glycyrriza glabra* L. (Leguminosae)	Licorice	Rhizome, root	Used as an expectorant, demulcent and flavor
12.	*Humulus lupulus* L. (Cannabinaceae)	Hops	Fruits, strobiles	Used as a sedative, sleep aid and mind-altering action
13.	*Hydrastis canadensis* L.$ (Ranunculaceae)	Golden seal	Rhizome, root	Treatment of bacterial and fungal infections of the mucous membrane, gastrointestinal tract infections that cause diarrhea, liver cirrhosis, inflamed gall bladder, eye infections, wounds and herpes labialis and as a prophylactic use for traveler's diarrhea
14.	*Hypericum perforatum* L.*$ (Hypericaceae)	St. John's wort	Leaf, tops	Treatments of mild to moderate depression, anxiety, antiviral, anti-bacterial and anti-inflammatory, wounds and burns, and blunt injuries
15.	*Linum usisttatissimum* L. (Linaceae)	Flaxseed	Seed	Treatment of chronic constipation, colons damaged by abuse of laxatives, irritable colon, divericultis, gastritis, enteritis and as a cataplasm for local inflammation
16.	*Liriosma ovata* Miers (Oleaceae)	Muira puama	Bark	For the treatment of dysentery and impotence
17.	*Paeonia officinalis* L. emend. Willdenow (Paeoniaceae)	Peony	Root	Antispasmodic, sedative, suppositories and whooping cough

Table I
Continued.

No.	*Botanical names (family)*	*Common name*	*Plant part consumed*	*Clinical indications*
18.	*Passiflora incarnata* L. (Passifloraceae)	Passion flower	Whole plant, flowering and fruiting top	To induce normal sleep with easy light breathing and little to no mental depression. Commonly used for relaxation and sleep
19.	*Pausinystalia yohimbe* (K. Schumann) Pierre ex Beille (Rubiaceae)	Yohimbe	Bark	Aphrodisiac, sexual stimulant and mild hallucinogen
20.	*Piper methysticum* G. Forster*$ (Piperaceae)	Kava-kava	Rhizome, root	Treatment of nervousness and insomnia, CNS depressant activity and euphoria, sleep inducement and anxiety reduction and skeletal muscle relaxant
21.	*Pueraria lobata* (Willd.) Ohwi (Linaceae)	Kudzu	Root	Treatment of alcohol dependence as an alcohol aversion therapy
22.	*Pygeum africanum* Hook.f.* (Rosaceae)	Pygeum	Bark, fruits, fruit trichomes	Treatment of prostatic hyperplasia particularly benign prostatic hypertrophy
23.	*Serenoa serrulata* Hook.f.*$ (Palmea)	Saw palmetto	Fruit	Effective in the treatment of benign prostate hyperplasia, prostate complaints and irritable bladder
24.	*Tanacetum parthenium* (L.) Schultz Bip* (Asteraceae)	Feverfew	Leaf, aerial parts, fruit, root, whole plant	Treatment of migraine, tinnitus, vertigo, arthritis, fever, menstrual disorders, difficulty during labor, stomachache, toothache and insect bites
25.	*Tribulus terestris* L. (Zygophyllaceae)	Tribulus	Fruit, root, whole plant	Treatment of impotence, kidney stones and diseases of the genito-urinary system such as gonorrhea
26.	*Turnera aphrodisiaca* Urb. (Turneraceae)	Damiana	Leaf, stem	As an aphrodisiac for coital insufficiency, depression, nervous dyspepsia, mild purgative, stomachic and atonic constipation
27.	*Uncaria tomentosa* (Willd.) DC. (Rubiaceae)	Cat's claw	Leaf, twig, stem, thorn	Used as a sedative, antispasmodic, liver protectant and to lower blood pressure
28.	*Urtica dioica* L.* (Urticaceae)	Nettle	Over-ground plant, root, bark	Anti-hemorrhagic and hypoglycemic effects. Also for the treatment of eczema
29.	*Vaccinum myrttilus* L. (Ericaceae)	Blueberry	Fruit, leaf	Used as a mild laxative, antibacterial and treatment of varicose veins, hemorrhoids and capillary fragility. Also used for the treatment of diarrhea and mild enteritis
30.	*Valeriana officinalis* L.*$ (Valerianaceae)	Valerian	Roots, rhizome	Used as a sedative, treatment of nervous tension during pre-menstrual syndrome, muscle spasm, menopause and restless motor syndromes
31.	*Vitex angus-castus* L. (Verbenaceae)	Chasteberry	Fruit	Treatment of insufficient lactation and menstrual problem resulting from corpus luteum deficiency including premenstrual symptoms
32.	*Zingiber officinale* Roscoe* (Zingiberaceae)	Ginger	Rhizome	Treatment of indigestion and motion sickness

∗ = Uses of medicinal plants supported by clinical data in WHO monographs.
$ = Among the 13 top selling herbs in the United States (1998).

Table II also shows the activity profile of all the extracts in the assay. Eighteen of these extracts inhibited the early antigen activation by more than 90% at a concentration of 100 μg ml^{-1}. The extract of turmeric exhibited the most active anti-EBV-EA activity at 10 μg ml^{-1} concentration while the extract of damiana herb showed the least activity at all the concentrations tested. However the possibility of further inhibitory effect cannot be ruled out at higher concentration. In the present study, we included eight of the 13 top selling herbs in the US market [17] and found the activity to be in the following order: kava-kava > black cohosh > echinacea > gingko

Table II
Inhibition of TPA induced EBV-EA activation by herbal extracts

			% Relative to control (% viability) Herbal extract (HE) concentration (μg ml^{-1} ratio/TPA)[a] TPA 32 pM = 100%		
No.	*Common herb name*[bc]	*Code [Yield %*[d]*]*	*HE conc. 100 μg ml^{-1}*	*HE conc. 10 μg ml^{-1}*	*HE conc. 1 μg ml^{-1}*
1.	*Angelica officinalis*[b] [RT, RZ]	KAP-3 [11.3]	18.9 ± 0.9 (60)	66.1 ± 1.2	100 ± 0.3
2.	*Asparagus officinalis* [RT, RZ]	KAP-34 [13.2]	26.2 ± 0.5 (70)	63.7 ± 2.0	100 ± 0.1
3.	*Boswellia serrata* [RE]	KAP-29 [CE]	6.4 ± 0.8 (70)	60.0 ± 2.1	92.7 ± 0.7
4.	*Caulophyllum thalictroides*[b] [RT, RZ]	KAP-38 [9.5]	9.7 ± 1.5 (60)	51.6 ± 2.0	98.1 ± 1.3
5.	*Cimicifuga racemosa*[b] [RT, RZ]	KAP-16 [7.5]	8.0 ± 0.5 (60)	53.2 ± 1.4	94.8 ± 0.2
6.	*Commiphora mukul*[b] [RE]	KAP-32 [1.6]	9.2 ± 0.7 (60)	51.3 ± 1.6	100 ± 0.4
7.	*Commiphora mukul*[b] [RE]	KAP-30 [17.7]	11.6 ± 0.5 (60)	54.3 ± 1.8	100 ± 0.2
8.	*Commiphora mukul*[b] [RE]	KAP-31 [4.0]	15.2 ± 0.3 (60)	59.6 ± 1.5	100 ± 0.9
9.	*Curcuma longa* [RZ]	KAL-1 [CE]	0.0 ± 0.0 (20)	0.0 ± 0.0 (60)	67.2 ± 0.4
10.	*Dioscorea villosa*[b] [RT, RZ]	KAP-4 [13.4]	6.6 ± 0.1 (70)	50.5 ± 1.2	93.9 ± 0.4
11.	*Echinaceae purpurea* [RT, RZ]	KAP-25 [2.3]	9.1 ± 0.1 (70)	60.8 ± 1.2	100 ± 0.1
12.	*Ginkgo biloba* [LF]	KAP-17 [17.9]	9.4 ± 0.3 (60)	55.7 ± 1.8	95.4 ± 0.3
13.	*Glycyrrhiza glabra* [RT, RZ]	KAP-4 [6.8]	7.4 ± 0.6 (70)	52.9 ± 1.3	94.0 ± 0.2
14.	*Humulus lupulus* [ST]	KAP-36 [2.2]	8.6 ± 0.5 (70)	50.2 ± 1.2	96.5 ± 0.3
15.	*Hydrastis canadensis* [RT, RZ]	KAP-37 [8.5]	9.5 ± 0.9 (70)	52.7 ± 1.4	100 ± 0.6
16.	*Hypericum perforatum* [WP]	KAP-40 [5.6]	18.4 ± 0.7 (60)	61.8 ± 2.1	100 ± 0.8
17.	*Linum usistatissimum* [SD]	KAP-44 [18.6]	5.7 ± 0.4 (70)	44.7 ± 2.5	91.8 ± 1.3
18.	*Linum usistatissimum* [SD]	KAP-45 [3.1]	12.5 ± 1.0 (60)	63.6 ± 1.8	100 ± 1.0
19.	*Linum usistatissimum* [SD]	KAP-28 [4.7]	27.0 ± 0.9 (70)	81.3 ± 1.5	100 ± 0.9
20.	*Liriosma ovata*[b] [RT]	KAP-10 [1.2]	17.8 ± 0.3 (60)	65.9 ± 1.2	100 ± 0.3
21.	*Paeonia officinalis*[b] [RT]	KAP-26 [2.8]	33.9 ± 0.3 (60)	68.0 ± 2.1	100 ± 0.2
22.	*Passiflora incarnata*[b] [WP]	KAP-8 [11.1]	3.1 ± 0.4 (70)	62.7 ± 1.3	94.8 ± 0.3
23.	*Pausinystalia yohimbe*[b] [BK]	KAP-27 [8.0]	4.7 ± 0.7 (60)	44.0 ± 1.8	94.0 ± 0.3
24.	*Piper methysticum* [RT, RZ]	KAP-19 [2.2]	4.4 ± 0.7 (60)	52.0 ± 1.6	93.8 ± 0.2
25.	*Pueraria lobata*[b] [RT]	KAP-9 [4.1]	11.0 ± 0.6 (70)	53.7 ± 1.1	94.6 ± 0.2
26.	*Pygeum africanum* [BK]	KAP-35 [1.1]	14.7 ± 0.7 (60)	54.6 ± 2.1	100 ± 0.6
27.	*Serenoa serrulata* [SD]	KAP-20 [9.2]	11.7 ± 0.6 (70)	58.9 ± 1.6	100 ± 0.4
28.	*Tanacetum parthenium* [LF]	KAP-21 [6.4]	8.0 ± 0.3 (60)	54.6 ± 1.2	95.1 ± 0.3
29.	*Tribulus terrestris*[b] [RT]	KAP-2 [6.3]	16.4 ± 0.3 (60)	65.0 ± 1.2	100 ± 0.3
30.	*Turnera aphrodisiaca*[b] [LF]	KAP-7 [15.3]	38.8 ± 0.4 (60)	75.6 ± 2.3	100 ± 0.5
31.	*Uncaria tomentosa*[b] [BK]	KAP-6 [7.0]	12.3 ± 0.7 (60)	57.4 ± 2.0	100 ± 0.4
32.	*Urtica dioica* [LF]	KAP-24 [3.5]	5.9 ± 0.5 (70)	48.2 ± 1.7	92.1 ± 0.2
33.	*Vaccinum myrtilus*[b] [LF]	KAP-22 [16.4]	14.5 ± 0.6 (70)	52.8 ± 1.3	100 ± 0.3
34.	*Valeriana officinalis*[b] [RT, RZ]	KAP-39 [17.5]	11.2 ± 0.5 (70)	59.0 ± 2.1	100 ± 0.1
35.	*Vitex angus castus*[b] [FR]	KAP-23 [3.3]	8.0 ± 0.6 (70)	51.9 ± 1.5	93.9 ± 0.1
36.	*Zingiber officinale* [RZ]	KAP-33 [15.5]	7.0 ± 1.0 (70)	49.2 ± 2.4	93.7 ± 0.3

Parentheses are omitted for cells which show 100% viability; HE conc. [a] = Herbal extract concentration (μg ml^{-1}). At the HE of 100 μg ml^{-1}, the viability of Raji cells was equal to or greater than 60% except for KAL-1 in every case. At this HE concentration, EBV-EA induction values for all extracts were significantly different from the HE-free control value ($P < 0.05$).
[b] New herbs associated with anti-cancer properties for the first time.
[c] Plant part extracted: RT = root; RE = resin; RZ = rhizome; LF = leaf; ST = strobile; WP = whole plant; SD = seed; BK = bark; FR = fruit.
[d] Yield (%) refers to amount of herbal material extracted by the solvent ethanol from the weighed plant material; CE = commercially available extracts were extracted with ethyl acetate followed by extraction with ethanol (KAP-31) and methanol (KAP-32). Flaxseeds were further extracted with *n*-hexane (KAP-44) followed by extraction with ethanol (KAP-45).

> golden seal > valerian > saw palmetto > St. John's wort. In Table II, we show the activity profile of 16 herbs identified for the first time as cancer chemopreventive agents. In all of the experiments with the exception of turmeric at 10 μg ml^{-1} concentration, the cell viability was between 60 and 70%, indicating relatively low toxicity of the extracts used in the study.

DISCUSSION

The EBV-EA activation promoted by phorbol ester has been employed for identification of cancer chemopreventive agents effective in the promotion stage of carcinogenesis which involves at least two reversible stages, conversion and propagation [21]. In our assay, the phorbol ester, TPA is a potent inflammatory agent. It is known to initially stimulate the signal transduction pathway enzymes such as protein kinase C and phospholipase A2 followed by activation or stimulation of a number of enzyme systems including lipoxygenases and cyclooxygenases of the arachidonic cascade and ornithine decarboxylase. Hence, substances that are effective inhibitors in this model will be interfering with one or more of these phorbol-ester-promotion-related processes and may be useful as cancer chemopreventive agents. In the present work, we have shown the *in vitro* comparative cancer chemopreventive profile of 36 herbal extracts in the EBV-EA model. We also demonstrated the potentials of 16 new medicinal plants in cancer chemoprevention. To the best of our knowledge, ours is the first study reporting the anti-cancer potentials of these herbs (Table II). The rank order identified turmeric and passion flower as the two most potent inhibitors of the anti-tumor promoting stage in this model.

Our earlier studies and several others have demonstrated the significant chemopreventive ability of turmeric [7, 22] in both chemically induced and virus-induced tumors, which is in agreement with the present results. In patients suffering from submucous fibrosis, treatment with turmeric extract reduced the number of micronulceated cells both in exfoliated oral mucosal cells and in circulating lymphocytes [23]. The turmeric constituent, curcumin is also a potent inhibitor of the epidermal growth factor receptor signaling in both androgen-dependent and androgen-independent prostate cancer cells [24]. This effect proceeds via the down-regulation of EGF-R protein, inhibiting the intrinsic EGF-R tyrosine kinase activity and the ligand-induced activation. In human colon epithelial cells, curcumin has been shown to inhibit cyclooxygenase 2 (COX-2) induction by the colon tumor promoters involving the inhibition of NF-kappaB via the NIK/IKK signaling complex [25]. Curcumin has also been shown to promote apoptosis [26] and inhibit the platelet-activating factor and arachidonic-acid-mediated platelet aggregation through the inhibition of thromboxane formation and Ca^{2+} signaling [27].

This present data are also compatible with other reports where cancer inhibitory effects have been observed for extracts of flaxseed [28], nettle [29], licorice [30, 31] and ginger [32]. Humulon, one of the bitters in hops, has been shown to markedly inhibit TPA-promoted skin tumor formation in mice [33] while boswellic acids from boswellin inhibited the cellular growth of human leukemia HL-60 cells [34].

In a randomized, placebo-controlled clinical trial involving 44 men aged 45–80 with symptomatic benign prostatic hyperplasia (BPH), saw palmetto herbal blend improved the clinical parameters including epithelial contraction [35]. Saw palmetto also showed such effectiveness in 42 BPH-treated patients without side effects [36]. In a similar study, pygeum, renowned for the treatment of BPH in herbal medicine, inhibited the proliferation of rat prostatic fibroblast via the inhibition of growth factors [37]. In a randomized, double blind study in 209 patients with BPH, Chatelain *et al.* [38] showed that pygeum treatment was significantly effective with satisfactory safety profile. These results taken together showed a novel modality by which one can interfere with the cellular events in the prostate cancer cells and prevent it from progressing to its hormone-refractory state [24]. Similar beneficial effects including anti-clastogenic activity were observed in Chernobyl reactor accident victims with ginkgo extract [39], in which regression or complete disappearance of the clastogenic factors in the plasma of these high-risk individuals was observed after two months of treatment. Such observations may in part be responsible for the increased acceptance of herbal remedies globally since present day science has been able to validate some of the beneficial ethnomedicinal uses [12–14, 17].

In the present study, the inhibition of TPA-induced promotion of EBV-EA activation may be related to the biochemical and/or molecular effects of the herbs or their constituents on some cascades in the pathway. The extract of ginger has been shown to induce ornithine decarboxylase, cyclooxygenase and lipoxygenase activities in a dose-dependent manner [32]. Another herbal formulation also used in our study, feverfew, has been shown in migraine patients to result in a reduction of genetic damage expressed in terms of chromosomal aberrations, sister chromatid exchange and mutation in the Ames/Salmonella assays [40]. Feverfew is an inhibitor of mitogen-induced human peripheral blood mononuclear cell (PBMC) proliferation, interleukin 2- (IL-2-) induced uptake and prostaglandin 2 (PGE2) release by interleukin 1- (IL-1-) stimulated synovial cells [41]. It also inhibited the generation of thromboxane B2, leukotriene B4 and chemoattractant-stimulated rat peritoneal leukocytes and human polymorphonuclear leukocytes [42]. Thus the observed anti-tumor activities observed in the study might proceed via these various mechanisms [22] including induction of apoptosis [43], enhanced DNA repair and immunostimulation demonstrated by cat's claw extract [44].

A rational approach in the field of cancer chemoprevention believes that inhibition of tumor promotion is a better strategy than inhibition of tumor initiation because initiation is a short irreversible event, whereas promotion is a long and cumulative process that is reversible during the initial stage [45]. Bearing in mind that long-term targeting of the tumor promotion stage could be a useful strategy for the prevention of cancer [8, 46], we have shown the potential use of these herbs in the reduction of human malignancies. Since they have been in use in different herbal recipes (Table I), their acceptance for additional health benefits will be easier and of great use in human cancer control. However, in the present study, we have chosen ethanol extraction as a uniform approach for a comparative activity profile, although these plants are prepared in various media. Additional work will entail preparing specific herbs in different media including the specific mode(s) of its human intake. In conclusion, our studies confirm the effectiveness of the herbs already associated with malignant control while defining new and additional uses for the additional herbs in the war against cancer through the chemopreventive approach.

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